What is Distinction? A Relational Approach

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Abstract

This paper introduces the concept of network *distinction* and develops a relational metric to capture it. The metric is theoretically inspired by Bourdieu's theory of agonistic competition for distinction in social fields, in which other actors and their actions provide opportunities and constraints for improving social positioning relative to others. The basic idea behind the network metric of distinction is to subtract each actor's *autonomy* from their *constraint*, so that highly autonomous and less constrained actors have greater distinction. Autonomy is measured by the focal actor's Eigenvector Centrality score (indicating connections to well-placed others), and constraint is defined by the average Eigenvector Centrality score of each actor's neighbors in a hypothetical network in which the focal actor is removed (indicating how others would benefit from that actor's exit from the network). The resulting node-level measure, when properly scaled to have a natural zero point, yields a measure of *distinction*, in which the uniqueness of an actor's relations is more heavily rewarded than under the standard Eigenvector Centrality score. After exploring the mathematical properties of the new measure in a number of idealized network structures, we analyze two classic datasets, Padgett and Ansell's (**?**) Medici Network and Zachary's (**?**) Karate Club, to show that the observed actions of the agents involved are consistent with a pursuit of distinction. We close by outlining how the resulting distinction measure can be used to both forecast network transformations and anticipate competitive changes resulting from external network shocks.

1 Distinction Beyond Taste

Popular within cultural sociology, the concept of distinction has generated a rich body of literature aimed at explaining cultural differentiation within social fields (??). From this perspective, the pursuit of distinction, as the expression of an actor's status and uniqueness, is an end in itself, providing the basic motivation for action (?). The question of cultural distinction is often treated empirically as a matter of the deference given to symbolic forms of differentiation enacted by designated institutional authorities (??). In this last respect, the more encompassing notion of *social* distinction, implying an actor's embeddedness in a relational field composed of concrete ties to other actors (??), is often measured as the shadow counterpart of cultural distinction or collapsed into the concept of status more generally (?). This approach overlooks how network constraints imposed by other agents affect actors' capacity to be distinct. A generalized, measured ordering of social distinction, derived from a network of concrete social relations, would provide Bourdieu's conception with a broader methodological basis for empirical analysis.

1.1 What is Distinction?

Bourdieu was primarily interested in the competition among actors for positions within the hierarchy implied by the field (??). Bourdieu defined the field as a social space where actors operate within socially negotiated boundaries that arbitrate relevant competitive claims over position (?). In interplay with the field, individuals develop a habitus of embodied dispositions, expressed as cognitive ease with particular types of settings, objects, and performances, allowing them to navigate those settings more deftly and to perceive and anticipate, more assuredly and efficiently, the affordances and opportunities for advancement they provide. An actor's position in the field, in turn, influences their future habitus. Similarly, their habitus determines their future possibilities within the field and shapes the opportunities available to others. Actors compete for positions in the field through various forms of capital or resources socially agreed upon as desirable within the field (?), thereby facilitating the formation of connections that both order and reflect the field's hierarchy. The interaction between an individual's desirability and the uniqueness of their capital determines their place in the field's positional hierarchy.

Bourdieu (?) most famously used the conceptualization of distinction to explain the persistence of the relationship between class and cultural hierarchies. Cultural elites in France distinguished themselves from their inferiors by adopting consecrated practices, thereby setting themselves apart from the common. There is a strong correlation between those in the position of cultural and economic elites. Still, such a correlation is not guaranteed at the individual level, even if it is relevant at the field level. among those in positions of cultural and economic elite status

While largely a concept used by researchers of cultural practices, the pursuit of distinction is a general expression of the interaction between status and uniqueness within a field of positions (?). For example, while cultural practices may be a marker of difference, the absence of association with others outside the milieu is also a matter of agency within it, a question of the constraints imposed by the set of relationships in which the actor is embedded. For Bourdieu, the struggle for position in a field was a product of a field of action and a field of forces (**?**). The field of action refers to the agency of actors actively advancing their interests. In contrast, the field of forces describes the counteracting pressures faced by the other participants whose interests would be disrupted by the actor's positional transformations.

In relation to theory, network status is another name for *influence* (**?**), and uniqueness is another name for *autonomy* in a social environment (**?**). Translated into network terms, Bourdieu's field of action can be understood as the network's status hierarchy, while the field of struggle reflects the constraints neighbors impose on an actor. An actor with strong influence is well-connected to others who can be mobilized for effective action. An actor with high autonomy is not constrained by their pattern connections or forced into action by them (**?**). We often use context-specific words to denote position in the hierarchy and the ability to act within it. Nevertheless, we use "status" and "autonomy" below as general terms to describe the actors' ability to maneuver for better positioning against the level of constraint imposed by external forces, which usually means other actors and their relations.

In developing a broader understanding and measure of social distinction, the interplay between actor-level incentives and structural impediments to action becomes apparent. When we understand the underlying motivations that drive actors to proactively reshape their network connections to achieve more distinct positions, historical patterns of network change can be better accounted for and, in some cases, even predicted.

1.2 Position as the Relational Basis for a Distinction Measure

Bourdieu's separation of distinction into habitus and position is not a sociological problem in itself (**?**); rather, it poses the analytical challenge of how an actor's network position generates dispositions, and how these dispositions, in turn, create opportunities for network mobility. Both dispositions and positions are fundamentally relational outcomes (**?**); however, when developing a network measure of distinction, only the relational basis of position is typically considered as this greatly simplifies the already daunting measurement challenge.

However, to the extent that position and habitus reflect each other through their symbiotic connection, empirically capturing only one of the two components is sufficient to measure a social system's distinction hierarchy with confidence over a short time window. Actors should not hold a network position that does not align with their habitus, and their current habitus and social position determine their ability to radically change positions. If the transformation of habitus is a stepwise interplay with positional movement that unfolds over time, the measure becomes less informative if it captures networks over extended periods. However, even over long periods, this theoretical collapse, which permits a network-measured methodological shortcut, may still be adequate in a moderately stable social environment.

To establish a firm theoretical foundation for the measurement of distinction, it is essential first to make explicit the relationship between empirical data characteristic of the social network analysis tradition. Network measures are fundamentally based on the idea that researchers capture relationships between pairs of agents.[1] A tie between two nodes captures, in some way, their connection, however it is defined. Such connections can be broadly classified as undirected when dyadic equality is implied, or directed when we need to track unequal flows, sentiments, or perceptions across both sides of the ledger. In the absence of direct measurement of directed elements, however, many networks are measured as composed of ties lacking directionality. However, even in networks composed solely of undirected relations, hierarchy can still be assessed from the pattern of relationships. In such networks, we can still rank the direction of power and deference through the *absence* of ties between nodes at different levels of the hierarchical structure. For example, in an organizational hierarchy, we can identify the CEO by observing relationships with vice presidents and the absence of direct ties to middle management and production workers. In this sense, hierarchical differences between nodes are mapped to a scalar index of *status* or *prestige* (**?**).

The first aspect of social distinction is, therefore, *status*. Status refers to the internal stratification of all network members, along with the connections of deference and obligation. In politics, status refers to the ability to access those with greater access, with those most central to the access network having the authority to make decisions. This can be captured traditionally by measures of status such as eigenvector centrality (**?**). Status is higher for actors when the people they are connected to are themselves influential, and accessibility is merely a political reframe of the context. The internal stratification of individuals is determined by who has stronger connections to those with even stronger connections, with those having few connections to other important people at the bottom of the hierarchy (**?**).

The second aspect of social distinction is *autonomy*, defined as the absence of network constraints (**?**). Being structurally unique minimizes the social constraints placed on an individual. The tighter the group's internal connections, the more they constrain any one individual in the group (**?**). This aspect of distinction takes the form of *constraint*. When an actor is one of many densely interconnected individuals, their ability to act contrary to their group is significantly weakened. However, when one is only a weakly dependent and connected member of a group, they have a greater ability to operate independently. Their ability to act is a source of distinction, as they can act independently. The question of structural uniqueness and autonomy is the same underlying consideration as captured by measures of path centrality. The more critical one is to bridging connections between individuals within a network, the greater one's importance and the lower one's constraint (**?**).

[1] Setting aside, for the sake of simplicity, more general representational formats such as hypergraphs and simplicial complexes, which can capture n-ary relations among arbitrary subsets of agents at a time (**?**).

?: 1263-4 were the first to bring attention to how the cohesion of those around a broker can structure the intermediary's own decisions. Cosimo de Medici, acting as a broker between two antagonistic groups, was constrained in his actions because decisive steps favoring one side would upset the other. However, this constraint was not an issue because Medici's interests lay in perpetuating his central role as broker, and, as the only source of interest, he was able to maintain his position. When conceptualizing distinction, the theory of robust action posits that those at the center of political networks, as long as their interests derive solely from maintaining that position, adopt moderate strategies that do not overly antagonize either side. The decisions are based on the need to ensure their continued central position in the network.

The interaction between influence and autonomy creates an actor's overall distinction in the network. The combination of influence and autonomy results in capacity, and the social reward for this capacity is distinction. An increase in one aspect can often, but not necessarily, lead to a decrease in the other. For example, a connection with someone who has high accessibility can reduce one's autonomy, and new access comes with new constraints. However, the trade-off can maximize one's overall distinction, yielding gains from the connection. Without such gains, the connection would not be formed.

2 A Generalized Measure of Network Distinction

In this section, we present our measure of distinction centrality, which incorporates a scalar adjustment factor to mitigate issues arising from arbitrary scaling. We also indicate that a custom *R* function is available to facilitate the implementation of the measure in future work.

2.1 Measuring Distinction

Formalizing the interaction between status and uniqueness allows for a measure of network distinction. As we have noted, distinction as a theoretical concept is composed of status mediated by autonomy. A distinction measure can reflect this by considering an actor's status and then discounting it by the counterfactual status of their neighbors when the focal actor is absent from the network. The more reliant an actor's neighbors are on the focal actor for their status, the less able the focal actor is to exert influence over them, and thus the lower the focal actor's distinction. In other words, autonomy is determined by how much one's neighbors rely on you to maintain their advantageous position in the network.

Accordingly, we propose the following as a measure of distinction:

$$\beta_i = \alpha s_i - \kappa_i \tag{1}$$

Where β_i is the distinction centrality score of actor i, s_i is actor i's status, κ_i is actor i's network constraint (the obverse of autonomy), and α is a scalar adjustment factor to be discussed below. Each individual's status score s_i is obtained from the vector $\mathbf{s}$ obtained from solving the standard linear eigenvector equation:

$$\lambda \mathbf{s} = \mathbf{A}\mathbf{s} \tag{2}$$

Where $\mathbf{A}$ is the network's adjacency matrix and $\mathbf{s}$ is the right dominant eigenvector of the matrix (**?**); status, therefore, is given by the usual eigenvector centrality score (**?**), taken as the absolute value of each individual's score on the first eigenvector of the adjacency matrix. When a node is an isolate, either in the larger graph or in the node-deleted subgraph used to compute constraint scores, we set its status score to $s_i = 0$.[2] When the graph is disconnected, the status vector $\mathbf{s}$ is assembled by summing the entries of the k eigenvectors of $\mathbf{A}$ where k is the number of connected components in the graph.

The second term κ_i in equation **??** is a *localized* version of an "induced centrality" score in the sense of **?**. According to **?**, an induced centrality is obtained by calculating the difference that removing a node from the graph makes to some *global* property of the graph (a graph invariant). **?** suggests taking the sum of the centrality scores of the nodes in the node-deleted subgraph produced by removing the focal node as a measure of that node's centrality, treating it as that node's contribution to the centrality of other nodes in the graph. They propose examples that extend this idea to classical centrality measures such as degree, closeness, and betweenness (**?**).

Here, we extend the induced centrality approach in two ways and incorporate it into our measurement of distinction. First, we use the induced centrality perspective in the context of "spectral ranking" (**?**). So far, in social network analysis, induced centralities have mainly been applied to non-local centrality measures aimed at capturing autonomy or brokerage, such as betweenness centrality.[3] Secondly, we look not at the difference that removing a node in the graph makes for a global property of the graph (e.g., the sum of the centralities of *all* other nodes), but by restricting it to its local consequences, namely, the difference it makes for the sum of the centralities of its *neighbors*.

Accordingly, the κ term in Equation **??** is given by the average status (eigenvector centrality) of i's neighbors in the *node-deleted subgraph* of the focal graph G that excludes i:

$$\kappa_i = \frac{\sum_{j \in N(i)} s_j^{G-i}}{|N(i)|} \tag{3}$$

Where $|N(i)|$ is the cardinality of node i's neighbor set (node i's degree), and s_j^{G-i}

[2]In the case of a network with no connections (such as an empty graph), some routines calculate it as $s_i = 1.0$. However, given that it is still a network with no connections, setting isolates to $s_i = 0$ is theoretically appropriate, since status reflects connections. It would be inappropriate for an isolate to have the same status as the highest-status node in a large network.

[3]With the exception of so-called "node vitality" measures of node position in network science, which are indeed induced centrality measures in the Everett-Borgatti sense based on matrix functions for counting all pairwise walks in a network.

is the status of node j in the graph that excludes node i.[4]

Given that both s_i and k_i are expressed in the (arbitrary) scaling produced by the eigenvector scoring procedure, it is desirable to rescale the distinction centrality measure so that it has a natural zero point. As equation **??** shows, we do this by multiplying each actor's status score by a scaling factor, which is given by:

$$\alpha = \frac{\bar{\kappa}}{\bar{\mathbf{s}}} \tag{4}$$

Where $\bar{\kappa}$ is the mean of the vector of κ_i constraint scores for each actor in the network, and $\bar{\mathbf{s}}$ is the mean of the vector of s_i status scores for each actor in the network. As hinted earlier, a desirable feature of rescaling the s_i component of equation **??** in this way is that in the case of a regular graph (e.g., a circle graph or a fully-connected clique), $\alpha s_i = \kappa_i$ and thus $\beta_i = 0$ for each node i (see Tables **??** and **??**).

Another advantage of rescaling the s_i component of equation **??** is that it makes the distinction score of key nodes in the graph sensitive to network size, so that it increases for the most distinct (central) node when peripheral actors are added to the network. For instance, Figure **??** illustrates the centrality score for central and peripheral nodes in star graphs of varying sizes. As we can see, when distinction is calculated simply as $\beta_i^* = s_i - \kappa_i$, the distinction score is a constant for the star nodes across graphs of different sizes, demonstrating insensitivity to network scaling. By way of contrast, when distinction is calculated as in Equation **??**, then we observe that the distinction score of the star node increases logarithmically with network size, which is precisely what we want, as a star mediating between many peripheral nodes should be more distinct that a star mediating between just a few peripheral nodes (we elaborate on the mathematical reasons for this below).

3 Distinction in Idealized Worlds

Below, we explore the properties of the proposed distinction measure by presenting distinction scores for a variety of toy models representing idealized yet informative network structures.[5] The function code is reproduced in the Appendix. In each model, there are seven nodes, and node 1 is the best-connected node in the network. Thus, node 1 always has an eigenvector centrality of 1. However, node 1's distinction varies with the interconnections among the other nodes, demonstrating how distinction centrality captures the extent of the constraint exerted by neighbors.

[4] The κ term is meant to capture the element of autonomy. This is substantively the same mechanism as in a jackknife resampling procedure, except that, rather than forming a sampling distribution, we measure local constraints. Thus, the κ_i component of equation **??** is larger for nodes whose neighbors would gain a lot in status (on average) if they were to be removed from the network.

[5] To conduct the analyses below, we use a custom function written for the *R* statistical computing environment (**?**), available at [url].

3.1 Maximum Inequality and Equality in Distinction

Figure **??** shows a star graph of order seven, in which node 1 is the central node connected to six peripheral nodes, each of which has no other connections. This hub-spoke system is the simplest way to explore the linkages among distinction, status, and constraint. The reason is that for the star node 1, κ_i is always zero. In other words, the star node has a minimum network constraint, given that the node-deleted subgraph obtained by removing the central node from the star graph is a disconnected, empty graph, where the status of the peripheral nodes collapses to zero, meaning that the star node's distinction score reduces to:

$$\beta_1 = \alpha s_1 \tag{5}$$

In a star graph, s_1 is a constant regardless of the graph's order (e.g., the number of peripheral nodes). This means that differences in the central nodes' distinction across star graphs of different orders are entirely driven by α. Note also that if we forgo scaling and set $\alpha = 1$ in Equation **??**, then distinction reduces to status ($\beta_1 = s_1$) and therefore the distinction of the star node fails to scale with increasing network size (see Figure **??**).

Further note that the average constraint of the star graph is also a constant, equal to $\left(s_1 - \frac{s_1}{N}\right)$, where N is the number of nodes in the graph. Substituting this expression for α in Equation **??** gives us the distinction score for the central node in the star graph:

$$\beta_1 = \frac{s_1^2 - \frac{s_1^2}{N}}{\bar{\mathbf{s}}} \tag{6}$$

This means that in the star graph, the central node's distinction is proportional to the square of its status score minus the same score divided by the network size, divided by the average status in the network, which declines (at a decreasing rate) as network size increases, as shown in Figure **??**. Thus, the distinction of the star node increases at a decreasing rate governed by the size of the graph N, as also shown in Figure **??**.

The hub of a star graph thus has the maximum possible distance between status and constraint. However, with additional spokes added to a star graph, the scalar-adjusted distinction measure increases. The hub of a star graph with seven spokes is more distinct than a hub with six, allowing for a degree of comparability across networks.[6] Distinction becomes a meaningful phenomenon only in networks of three or more nodes.

Similarly, for every peripheral node j, the constraint is constant and equal to the status score of the central node s_1. Thus, for the peripheral nodes in a star graph, Equation **??** becomes:

$$\beta_j = \alpha s_j - s_1 \tag{7}$$

Substituting Equation **??** into Equation **??** we obtain:

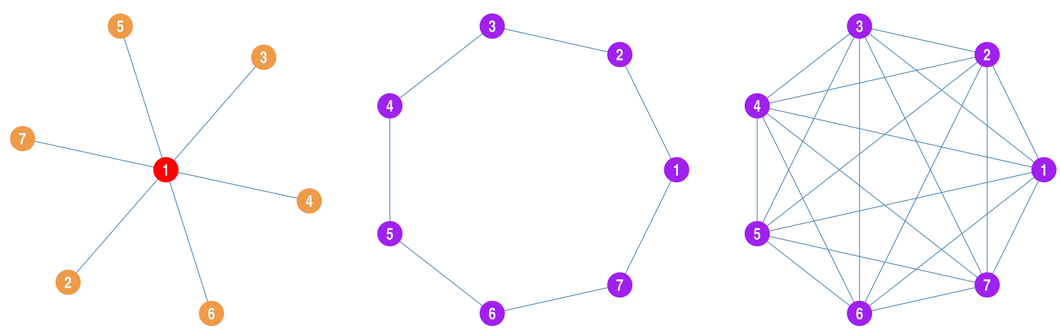
[6] Nodes in a graph composed of a single dyad would have a scalar-adjusted distinction of 0.00 because the average constraint is 0.

Table 1: Distinction centrality scores for Figure 1.

Nodes	β_i	s_i	κ_i	α
Star (1a)				
1*	1.739	1.000	0.000	1.739
2, 3, 4, 5, 6, 7	-0.290	0.408	1.000	1.739
Circle (1b)				
1, 2, 3, 4, 5, 6, 7	0.000	1.000	0.445	0.445
Clique (1c)				
1, 2, 3, 4, 5, 6, 7	0.000	1.000	1.000	1.000

$$\beta_j = \alpha s_j - \frac{\beta_1}{\alpha} \tag{8}$$

This tells us that the distinction of the peripheral node in the star graph is proportional to its own (scaled) status, discounted by the distinction of the star node divided by the same scaling factor.



(a) Star Graph (b) Circle Graph (c) Clique Graph

Figure 1: Distinction Centrality in toy graphs. Nodes with the highest distinction are shown in red, nodes with positive distinction are shown in blue, nodes with negative distinction are shown in tan, and nodes with distinction close to zero are shown in purple.

If the star graph is the ideal-typical representation of maximum inequality (and thus distinction), the circle and clique graphs (see Figures **??** and **??**) represent the obverse case; a representation of maximum equality and thus *lack* of distinction. In that respect, distinction should be at its theoretical minimum of zero for all nodes in these cases. As we can see in Tables **??** and **??**, this is precisely what we observe.

In the case of the circle graph, status scores are equivalent across all nodes ($s_i = 0.38$ for the circle graph of order seven shown in Figure **??**. Also note that removing a node from the circle graph yields a path graph of order $N - 1$, where N is the number of

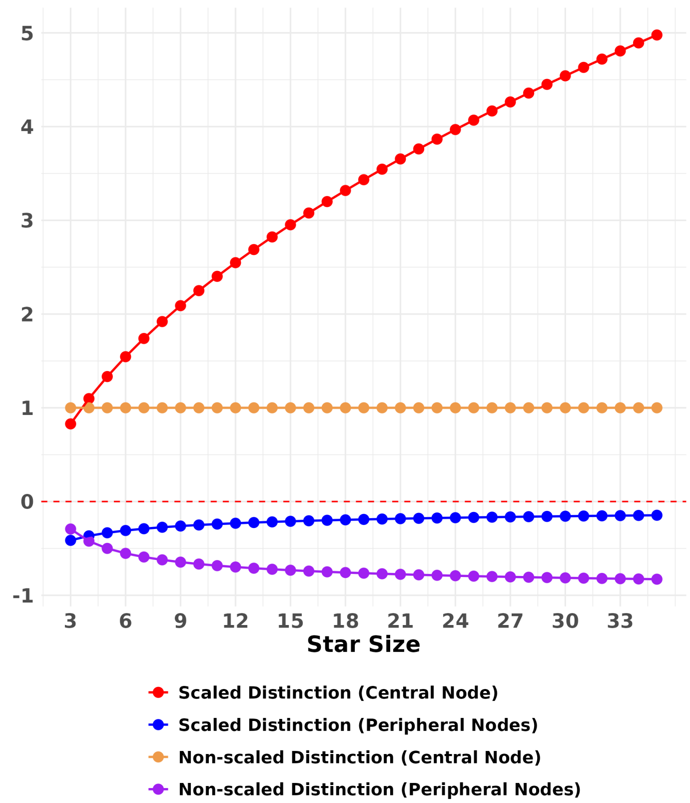


Figure 2: Scaled and non-scaled distinction centralities for the two node roles in the star graph.

nodes in the original circle graph. The previous neighbors of the removed node become the end nodes in the new path graph. These two nodes will, by necessity, have the same status score. This means that the κ component of Equation **??** will be the same constant across all nodes, and that the average status and average constraint will also be equal to a constant, in which case Equation **??** becomes:

$$\beta_i = \left(\frac{\kappa_i}{s_i} s_i - \kappa_i \right) = (\kappa_i - \kappa_i) = 0 \tag{9}$$

Thus, the distinction is zero for all nodes in the circle graph. The same reasoning applies to the clique graph, where nodes have the same constant status scores as in the circle graph. The difference is that in this case the κ component of Equation **??** is

Table 2: Distinction centrality scores for the circle graph.

Nodes	β_i	s_i	κ_i	α
1, 2, 3, 4, 5, 6, 7	0	1	0.445	0.445

Table 3: Distinction centrality scores for the clique graph.

Nodes	β_i	s_i	κ_i	α
1, 2, 3, 4, 5, 6, 7	0	1	1	1

obtained from the status scores of each actor in the node-deleted subgraph, excluding the focal node, which is just a clique of size $N-1$ where N is the number of nodes in the original clique. Just like in the circle graph case, however, both the s and κ components are constant across nodes, which means that the status scores cancel out by division and the constraint scores by subtraction, leaving everyone with a scaled distinction score of zero.[7]

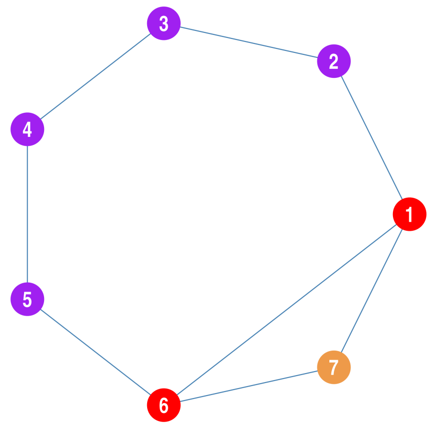
3.2 Distinction in Triadic Closure

Triadic closure is an observed regularity in which a connection is likely to form between two actors who, although not originally connected, are first connected to a common alter (Simmel 1908). In essence, friends of friends are more likely to become friends. The causes of this regularity and the reasons for its occasional failure are largely based on considerations of valence. Positive or negative affect drives the resulting triadic relations. The original consideration was the cognitive balance of the individual, or the psychological congruity of affect toward two objects they see as causally connected. As such, the relationships of all three actors become interdependent via their joint combination as "the parts of a causal unit" (**?**: 107).

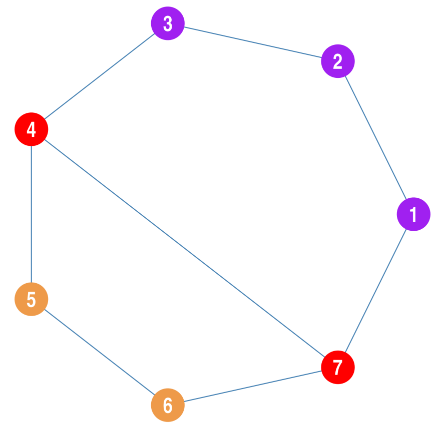
Beyond the cognitive congruence account of triadic closure, there is an accounting of triadic closure based on the strains new positive network connections may or may not create in existing relationships. Networks are sets of triads that are either actualized in connection, either positive or negative, or left unrealized. Actualization is largely predicated on propinquity, and the valence of theoretically possible ties never rises to the level of concern in a social question (**???**). However, in a contested context where ties are either positive or negative, triadic closure becomes a matter of not betraying one's friends by associating with one of their enemies, and of solidifying social alliances. However, triadic closure with a friend's friend signals commitment. The endorsement effect of triadic balance, that betrayal creates two relational problems rather than one, reinforces the surety of the actors (**?**).

These are all laudable explanations of the phenomenon with storied histories in

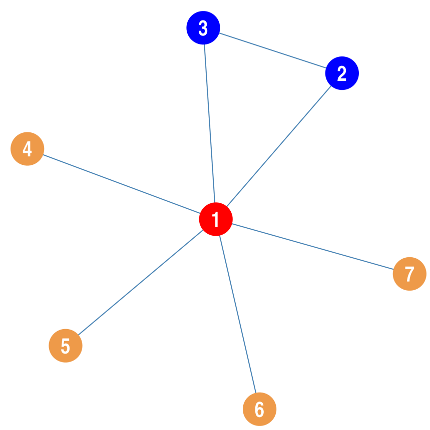
[7]Because of the local nature of the κ component in Equation **??**, in a regular (n, k) graph of order n with prescribed degree k across nodes, we should not expect scaled distinction to be exactly zero, since the focal nodes previous neighbors will have different status scores than the other nodes in the graph in the focal-node-deleted subgraph. However, we should not expect their status to be wildly different, since their degree is now only $k-1$. Instead, we should expect scaled distinction to fluctuate *around* the zero value across nodes. Nevertheless, these should cancel out across nodes; therefore, we should also expect the *average* distinction across nodes in a regular (n, k) graph to be zero.



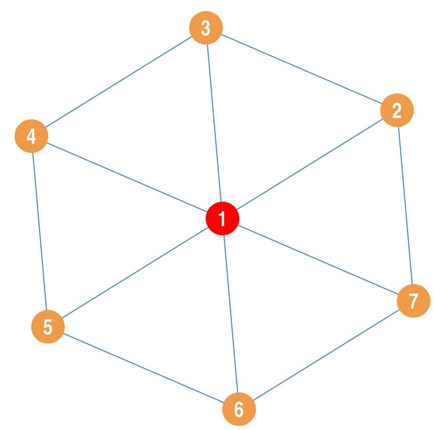
(a) Circle Graph with Closure



(b) Circle Graph with a Long Tie



(c) Star Graph with Closure



(d) Wheel Graph

Figure 3: Distinction Centrality in toy graphs. Nodes with the highest distinction are shown in red; nodes with positive distinction in blue; nodes with negative distinction in tan; and nodes with distinction close to zero in purple.

social network analysis. Yet we can also offer a distinction-based explanation of the relational incentives that motivate actors to engage in triadic closure. Figure 3 *Model (a)* and Table 4 present a graph and corresponding measures of near-neighbor triadic closure. The distinction of nodes 1 and 6 increases to .106, compared to 0 in the circle graph, while all other nodes fall below 0 due to the reverberating effects of the new connection. Notably, node 7 increases in status from the combination, but overall decreases in scaled distinction due to the tie. This example demonstrates the fragility of the circle graph as an equality condition in a distinction-based theoretical framework, as there are opportunities for mutual increases in distinction through triadic closure. While both status and distinction would predict tie formation in this simplified social

Table 4: Distinction centrality scores for Figure 3.

Nodes	β_i	s_i	κ_i	α
Circle with Near Neighbor Closure (2a)				
1, 6	0.209	1.000	0.564	0.773
2, 5	-0.030	0.586	0.483	0.773
3, 4	0.001	0.414	0.320	0.773
7	-0.360	0.828	1.000	0.773
Circle with Long Tie (2b)				
1, 3	-0.033	0.651	0.532	0.766
2	-0.025	0.548	0.445	0.766
4, 7	0.136	1.000	0.630	0.766
5, 6	-0.090	0.726	0.647	0.766
Star with One Triangle (2c)				
1*	1.237	1.000	0.333	1.57
2, 3	0.210	0.595	0.724	1.57
4, 5, 6, 7	-0.414	0.373	1.000	1.57
Star with Two Triangles (2d)				
1*	0.946	1.000	0.402	1.348
2, 3, 5, 6	0.028	0.539	0.699	1.348
4, 7	-0.528	0.350	1.000	1.348
2-Connected Subgraphs with Long Tie (2e)				
1, 7	0.025	0.654	0.701	1.11
2, 6	0.042	0.612	0.637	1.11
3, 5	-0.248	0.465	0.765	1.11
4*	0.361	1.000	0.749	1.11
Wheel (2f)				
1*	0.102	1.000	1.000	1.102
2, 3, 4, 5, 6, 7	-0.017	0.608	0.686	1.102

structure, the scaled distinction measure indicates that Node 7 would actually lose out in some respects under this arrangement. This matches findings that initial connections often weaken when triadic closure occurs (?), consistent with a reaction to the increased constraint placed on the closure broker. However, the actual incentives and consequences for each node in a triadic closure will depend on the overall network structure. Node 7 is worse off due to the structure of the circle graph, rather than because it will always be worse off under all conditions. If Node 7 is actually highly central to the graph, near-neighbor triadic closure can be distinguished beyond questions of balancing valence, as is the current framework for understanding triadic closure. A distinction-based theoretical framework can explain both the propensity towards triadic closure and its failure to form, depending on the initial patterning of relationships.

Figure 3 *Model (b)* and Table 5 present a graph and corresponding measures for a long-tie formation spanning the circle graph. The connected nodes 4 and 7 both have distinction-maximizing incentives to form a tie, but, notably, their scaled distinction increases more when they form a triad with a near neighbor. Thus, under sparse overall network conditions, the gains in status from local consolidation outweigh the gains in distinction from increased autonomy. Conversely, once dense local conditions prevail, long ties can become more valuable as the gain in autonomy outweighs the marginal gains in status. However, the unique topography of networks requires careful consideration in each case.

What the difference between these two examples suggests is that, based on considerations for distinction, the initial spread of social organization occurs locally through the triadic balance of near neighbors. This suggests that the natural form of organizational formation is a pyramid composed of combinations of local triads, further connected by brokers or welds that create structural folds. However, the transitivity created by the initial triad formation would spread quickly throughout the network, making welding structural folds a highly advantageous strategy for members of triads who could facilitate that connection, which is further explored below. Additionally, the example of triadic closure shows that only mild additions to network ties can very quickly create stratification. The addition of a single connection between nodes 1 and 6 in Figure 3 *Model (a)* stratified the network from an equal state into four hierarchical levels.

3.2.1 Distinction and Brokerage

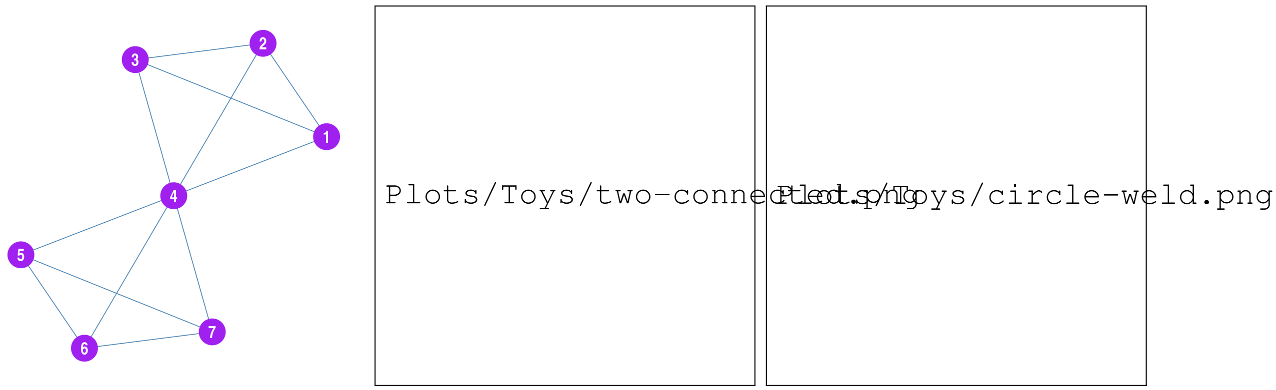
Model (b) presents an idealized depiction of a structural fold (**?**). The resulting measurement scores are in Table 2. While node 1 maintains a high status score, the status of the other nodes increases, and their constraints decrease. It is actually the constraint of Node 1 that increases, reaching 1.00 because all its neighbors are maximally interconnected in their respective subgroups. This brokerage position is more constrained than in the star graph because the other actors are connected in subgroups. The actors in the subgroup can combine their interests to counterbalance Node 1's central independence. In an unadjusted measure of distinction, the different nodes become remarkably close in their distinction measure. With a scalar adjustment, stratification increases, giving Node 1 a slight advantage while also highlighting the extent to which Node 1's distinction is eroded. In a maximally connected graph, each node's distinction would also go to 0.00, as there is no network differentiation among nodes.

Model (c) presents a graph that is 2-connected in the absence of node 1. Each subgraph is one connection short of forming complete subgraphs in the absence of Node 1. Compared to the structural fold, connections 3-2 and 6-7 have been removed. Measurement results are in Table 3. Of the demonstrative networks here, this is the one that most approximates a three-tiered ordered hierarchy. The resulting scalar-adjusted distinction measures well reflect this stratification, with Node 1 retaining its position, Nodes 4 and 5 occupying the middle, and Nodes 2, 3, 6, and 7 falling to the bottom.

Model (d) is the same 2-connected graph with the addition of a long tie connecting nodes 2 and 7. This long tie stratifies the network even more than in Model (c), bringing

Table 5: Distinction centrality scores for Figure 4.

Nodes	β_i	s_i	κ_i	α
Structural Fold (4-cliques) (3a)				
4*	-0.004	0.608	0.628	1.026
1, 2, 3, 5, 6, 7	0.026	1.000	1.000	1.026
Structural Fold (5-cliques) (3b)				
5*	0.003	0.588	0.573	0.98
1, 2, 3, 4, 6, 7, 8, 9	-0.020	1.000	1.000	0.98
Two-Connected, Welded Folds (3c)				
1, 3, 5, 7	-0.170	0.500	0.757	1.174
2, 6	0.155	0.618	0.570	1.174
4*	0.369	1.000	0.805	1.174
Welded Circles (3d)				
1, 3, 5, 7	-0.067	0.612	0.634	0.925
2, 6	0.026	0.500	0.437	0.925
4*	0.218	1.000	0.707	0.925
Intermediated 3-cliques (3e)				
1, 6	0.209	1.000	0.564	0.773
2, 5	-0.030	0.586	0.483	0.773
3, 4	0.001	0.414	0.320	0.773
7*	-0.360	0.828	1.000	0.773
Intermediated 4-cliques (3f)				
1, 5	-0.392	1.000	0.824	0.432
2, 3, 4, 6, 7, 8	0.252	0.849	0.115	0.432
9*	-0.728	0.629	1.000	0.432
Tree with 2 Branches, 2 Leaves (3g)				
1, 4	0.579	1.0	0.192	0.771
2, 3, 5, 6	-0.232	0.5	0.618	0.771
7*	-0.229	1.0	1.000	0.771
Tree with 2 Branches, 3 Leaves (3h)				
1, 5	0.764	1.000	0.125	0.889
2, 3, 4, 6, 7, 8	-0.220	0.447	0.618	0.889
9*	-0.205	0.894	1.000	0.889
Tree with 2 Branches, 2 Leaves, 1 Long Tie (3i)				
1, 4	0.420	1.000	0.318	0.739
2, 6	0.045	0.834	0.572	0.739
3, 5	-0.400	0.455	0.736	0.739
7*	-0.130	0.910	0.802	0.739
Tree with 2 Branches, 3 Leaves, 1 Long Tie (3j)				
1, 5	0.708	1.000	0.196	0.904
2, 7	0.108	0.719	0.542	0.904
3, 4, 6, 8	-0.347	0.418	0.726	0.904
9*	-0.244	0.836	1.000	0.904



(a) Structural Fold (4-Clique) (b) Two-Connected (c) Circle Weld

Figure 4: Comparison of betweenness brokerage and a structural folds Distinction Centrality in toy graphs. Nodes with the highest distinction are shown in red, nodes with positive distinction are shown in blue, nodes with negative distinction are shown in tan, and nodes with distinction close to zero are shown in purple.

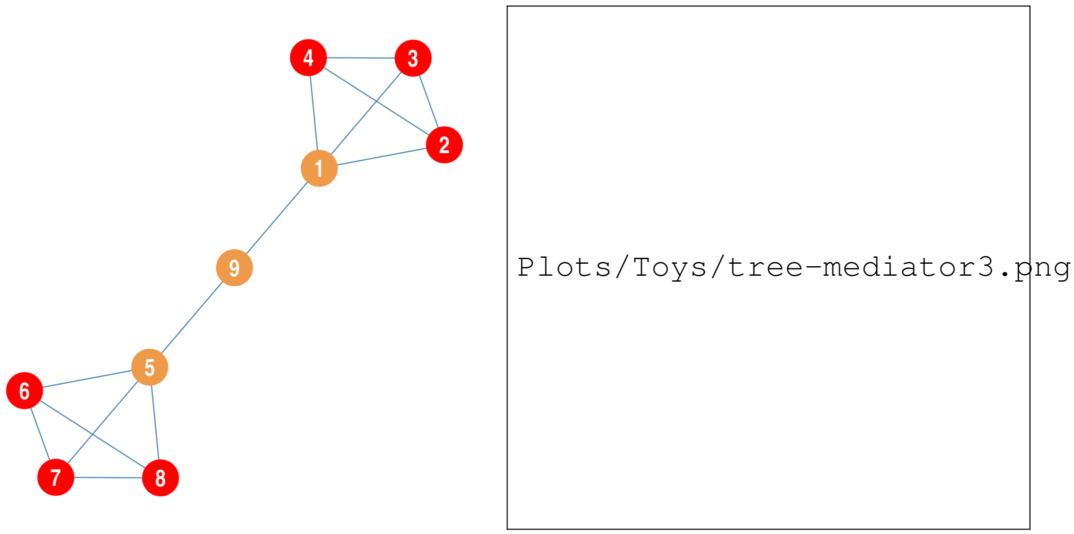
Nodes 2 and 7 quite close to 4 and 5. The formation of the long tie served to bolster the individual distinction of 2 and 7 more than the formation of the tie that would create the completely connected subgraphs, as in Model (b). In fact, all nodes except 3 and 6 have greater distinction in this case. This reflects the positional idea that distinction is a matter of stratification, and cohesion is at odds with creating stratification. While the status of Nodes 2 and 7 actually increases beyond that of Nodes 4 and 5 due to the long tie, the constraint also increases as they are now both further constrained by the influence of Node 1, their mutual neighbor.

4 Data Examples

4.1 Betweenness vs Distinction: Revisiting the Medici Network

The interplay of influence and autonomy creates an actor's overall distinction within the network. The combination of influence and autonomy results in capacity, and the social reward for this capacity is distinction. An increase in one aspect can often, but not necessarily, lead to a decrease in the other. For example, a connection with someone with high accessibility can lead to a decrease in one's autonomy, and new access comes with new constraints. However, the trade-off can maximize one's overall distinction, yielding gains from the connection. Without such gains, the connection would not be formed.

One of the consecrated papers of social network analysis is Padgett and Ansell's (1993) analysis of the centrality of the Medici to Florentine elite marriage and business



(a) Brokerage Mediated (4-Cliques) (b) Tree with 2 Branches, 3 Leaves

Figure 5: Comparison of betweenness brokerage and a structural folds Distinction Centrality in toy graphs. Nodes with the highest distinction are shown in red, nodes with positive distinction are shown in blue, nodes with negative distinction are shown in tan, and nodes with distinction close to zero are shown in purple.

networks in the 15th century. The empirics led to the theory of robust action, demonstrating that one's position in the network can serve as the basis for one's interests and that by balancing competitors, an actor can amplify their social position.

A puzzle that emerges from the empirics, however, is how the balance between antagonistic factions created by the Medici was undone within a decade of Padgett and Ansell's focus. The brokerage created social stability, yet in 1432, Rinaldo Albizzi, as leader of a coalition of conservative old oligarchic families, seized the reins of the Florentine government, exiling the leading Medici from Florence. Why was the Medici's network position as brokers so unstable? In other words, what were the network incentives of the old conservative families to attempt to remove the Medici from their network position?

The opportunities for increased social distinction explain the conservative oligarchs' attempt to remove the Medici from the network. Padgett and Ansell built their network of Florentine marriages and economic exchange as a series of blockmodels, which capture the inter-relations among families (**?**) 1276. Underneath the blockmodel are even more families in their Appendix B, for a total of 92 elite families considered (1276). The politics of Florence centered around family alliances consecrated through marriage, trade, and friendship ties. However, over the years, many families had developed cadet branches, with many old families having 10 or more households, with the Strozzi and Bardi having more than 50. Newer families had smaller, less ancient family networks, and so inter-family political cooperation was important for their social protection (150-

1). Family capacity, measured by size and wealth, was then channeled into inter-elite marriages, creating the networks of power and influence through which Florentine politics operated. The unity of the households within a single lineage was variable however, often depending on their own marriages. For example, the Bardi family largely sided with the old families against the Medici, but the marriage of Contessina de Bardi to Cosimo de Medici (129) meant that the family was not united in opposing the Medici.

To empirically examine the opportunity to gain social distinction in the Florentine network, Figure 2a from Padgett and Ansell was reproduced as an edgelist, with each tie between blocks counted as a single connection, regardless of its quality or frequency in the original. It is essentially a simplified network of blocks, treated as network actors, that are socially connected, indicated by the family names originally used by Padgett and Ansell. Ignoring the intensity of ties is unlikely to change the results given that most are singly connected in the original figure, and the multiple ties are still fragmented among a variety of households within a family, so they could inordinately strengthen connections rather than explain fracturing that occurred within families. Examining the Florentine network with the above measure of social distinction explains the transformation from political stability to fragmentation as a result of actor-level network incentives.

Results are presented in Table X. In the original state of the network, the Medici had the highest betweenness, while the Peruzzi had the highest status. The Albizzi, with the second-highest status, owed their high status in part to their connection with the Medici. Without the marriage connection to the Medici, the Albizzi would still have high status in the networks, but would otherwise be eclipsed by the Guasconi, who were also connected to the Medici and other major families. The Medici were actually of relatively middling status in the network, clearly deriving their power from the brokerage leadership.

In a comparison of the network with the Medici and the one without them, one notices that the status of the oligarchic families does not automatically increase. While those further away from the Medici, such as the Strozzi and Panciatichi, would see their status increase in the Medici's absence, as the center of the network would shift towards them, the Guasconi and the Albizzi would lose by this straightforward measure of status change. Considering the matter of constraint, the Guasconi and Albizzi would also become more constrained in the Medici's absence from the network. Thus, even as their status declined and their constraints increased, these families still chose to act against the Medici!

The status and distinction of the Guasconi and Albizzi would still be lower without the Medici in the network than with them. However, when adjusting for the levels of status and constraint in the system, the Guasconi and Albizzi, along with the rest of the conservative oligarchs, would increase their social distinction. The overall decrease in average status within the system, along with an increase in constraint, would make the status that the Guasconi and Albizzi did have at the level of the system increasingly important. The inequality within the system would essentially increase without the presence of the Medici, and ending up in the elite cluster of a more unequal network of

Plots/medici.png

(a) Medici Network

Plots/nomedici.png

Table 6

Node	Distinction	SDistinction	Status	Constraint	Scalar
Medici	-0.040	0.672	0.576	0.615	2.237
Guicciardini	-0.046	0.236	0.228	0.275	2.237
Ginori	-0.697	-0.515	0.147	0.844	2.237
Tornabuoni	-0.158	0.054	0.172	0.330	2.237
Albizzi	0.330	1.496	0.943	0.613	2.237
Guasconi	0.167	1.257	0.882	0.715	2.237
Cocco-Donati	-0.628	-0.456	0.139	0.768	2.237
Valori	-0.100	-0.060	0.032	0.131	2.237
Dietisalvi	-0.105	-0.063	0.034	0.139	2.237
Dal'Antella	-0.105	-0.063	0.034	0.139	2.237
Davanzati	-1.000	-1.000	0.000	1.000	2.237
Orlandini	-1.000	-1.000	0.000	1.000	2.237
Guadagni	-1.000	-1.000	0.000	1.000	2.237
Fioravanti	-1.000	-1.000	0.000	1.000	2.237
Bishceri	-1.000	-1.000	0.000	1.000	2.237
Ardinghello	-0.153	-0.092	0.049	0.202	2.237
Da Uzzano	-0.705	-0.441	0.213	0.918	2.237
Benizzi	-0.073	0.310	0.310	0.383	2.237
Baroncelli	-0.103	0.034	0.111	0.214	2.237
Velluti	-0.391	-0.075	0.255	0.646	2.237
Lamberteschi	-0.182	-0.109	0.058	0.240	2.237
Solosmei	-0.165	0.049	0.174	0.339	2.237
Della Casa	0.002	0.727	0.586	0.583	2.237
Altoviti	-0.680	-0.413	0.216	0.896	2.237
Peruzzi	0.327	1.564	1.000	0.673	2.237
Panciatichi	-0.251	0.491	0.600	0.851	2.237
Strozzi	0.008	0.844	0.675	0.667	2.237
Aldobrandini	-0.771	-0.487	0.229	1.000	2.237
Castellani	-0.257	0.297	0.448	0.705	2.237
Rucellai	-0.486	-0.295	0.155	0.641	2.237
Rondinelli	-0.121	0.422	0.439	0.560	2.237
Pepi	-0.319	-0.192	0.103	0.422	2.237
Scambrilla	-0.319	-0.192	0.103	0.422	2.237

elites was an increase in social distinction sufficient to entice members of the Guasconi and Albizzi blocks. In contrast to the application of an edge cutting algorithm that may shear the Guasconi and Albizzi blocks from the Medici due to the weight of their connections with the rest of the network, the members of the families in these blocks had distinction to be gained, as long as their own personal connections did not otherwise tie them to the Medici.

It was not just Cosimo de Medici who derived his sphinxlike qualities from his position in the network, but Rinaldo degli Albizzi was likewise noted for his unwillingness to take strong stances on behalf of the conservative faction of old elite families (Kent 1978:289). Given Rinaldo's similarity to Cosimo's brokerage network position, the structural incentives towards equivocation were similarly expressed. However, Rinaldo's structural opportunities were transformed with the death of Niccolò da Uzzano, the previous leader of the conservative faction. While the Uzzanos are attached to the Guasconi block in the original graph of relations, they actually shared deep family ties with the Bardis, who fell into the same block as the Guasconi. Per Table B1 in Padgett and Ansell (1314-6), the Bardi were third in gross family wealth after the Strozzi and Medici. Niccolò's mother was a Bardi, and his brother married into the family (Kent 1978:166), elevating Niccolò's authority beyond what might otherwise be discernible from the original graph. The Uzzano and Bardi leadership of the conservatives balanced the old elites away from Rinaldo, making ties to the Medici more important for their network position. Niccolò was also seen largely as moderate despite leading the old families against the new families (Kent 1978:256). However, Niccolò's death in 1432 created a gap in the conservative faction's leadership for Rinaldo to fill. The center of power within the network of old families was renegotiable, and in exchange for eschewing the Medici, the Rinaldo degli Albizzi could seize leadership and shift the network's emphasis towards himself. The secondhand data from Padgett and Ansell's block models does not allow for a clear demonstration of the movement, but the historiography would indicate the underlying shifts in network opportunity.

With Bernardo Guadagni's rise to the Gonfaloniere, Albizzi and the old families moved to exile Cosimo and Lorenzo de Medici, seemingly to seize the levers of city government through factional decapitation, thereby avoiding destructive conflict (289-90). While the conservatives seized government, the Medicans were not undone by the loss of their leaders but rather resisted without prompting a crisis (299). The moment of crisis came with the lottery, whereby chance, the Medici supporters came to dominate the Priors (328). The resulting re-entry of the Medici into government power threatened the conservative faction, and Rinaldo responded by mobilizing the forces of the conservative families to seize the castle where the priors were meeting. The Mediceans gained word and were able to reinforce the castle before the conservatives could arrive. In this moment of constitutional crisis, Palla Strozzi refused to commit his large force of support to Rinaldo's cause, and even the Peruzzi demobilized upon receipt of a deal from the Medicean's (331-4). The crisis itself was negotiated down through the intervention of the Pope, but the crisis shifted momentum to the Medicean's side through the fragmentation of the conservatives and the outpouring of armed support flowing in from outside the city walls (Kent 1978:330-4). The leadership of the old families was subsequently undone for their crimes (Kent 1978:335-48). The conservatives would clearly improve their distinction within Florence with the removal of the Medici, yet in the end, the elite network does not explain their defeat. The threatened major social losses to the newer families explain their dogged resistance to the conservative ascension, yet the support for the Medici with the Pope and beyond Florence was critical as the balance of force among the elite families was in the conservatives' favor (see Padgett and Ansell Appendix B).

4.2 Group Detection vs Distinction: Revisiting the Karate Club

Another consecrated paper in network analysis is Zachary's (1977) examination of the fissure of a university karate club network. Zachary found that the tension in the group centered around questions of compensation for the outside karate teacher with the internal student group leadership. However, Zachary finds that the dispute was channeled and resolved along an already existing division in the underlying friendship connections among the club members (1977:454-5).

The paper was an early application of an algorithm to predict the lines along which the group would divide. The original Ford-Fulkerson Labeling algorithm was a network flow model that sought to use node distances to detect community membership (1977:454). However, in the original algorithm, Node 9 was misclassified as belonging to the Officer's network, whereas empirically, Node 9 joined the network of the karate teacher, Mr. Hi. The original discrepancy was explained by Node 9 being near achieving a black belt, so the institutional opportunity costs outweighed the social network costs.

Contrary to expectations, the application of the distinction centrality measure did not explain why Node 9 left for Mr. Hi's network. The original hypothesis was that, because the Officer's network closely resembles a star graph, Node 9 would command less distinction in the network than in Mr. Hi's more heterogeneous network. Results are presented in Table X. The measure indicates that Node 9 would have had a greater overall relative distinction if he had remained in the Officer's network. The idea being that, in Mr. Hi's network, Node 9 would be more constrained than in the officer's network, but would have higher status because he was directly connected to the high-status center in an otherwise stratified network.

The exercise, however, focuses attention on the boundaries of the field. The original explanation for the discrepancy points to Node 9's large karate capital. If the field is bounded as the "karate club," Node 9's decision is unexpected, but if the field is "karate practitioners," much of the field might go unobserved in Zachary's original study. We do know that some of the students, including Node 9, practiced at Mr. Hi's dojo as well as the club. The fact that Node 9 had club connections only to the most central members of both the officer's and Mr. Hi's networks would indicate that Node 9 only affiliated with likely similarly advanced members of the club. If Node 9 were similarly affiliating with advanced students who only attended the dojo, then this would likely influence the decision regarding distinction at the field level. Failure to fully capture positions within the appropriately defined field can result in measures that fail to accurately reflect empirical outcomes.

5 Distinction as Search Algorithm

Given the value that eigenvector-style centrality measures have had in underpinning the logic of internet search algorithms (???), the application of distinction centrality to a reference network is briefly considered. As the social logic of distinction centrality involves jointly considering the interaction between status and constraint, in a reference network the distinction measure will give greater priority to the uniqueness of a node's flow of affiliations.

A toy reference network is presented in Figure 3. The network was designed to contrast status and distinction within a graph, featuring a clear cluster of nodes with high density and tenuously connected to a node with a high degree of popularity external to the main, denser cluster. Such a graph is supposed to represent a contrast between an orthodox set of mutual endorsements and the emergence of a heterodox set of connections.

The resulting distinction scores are presented in Table 6. The results are ranked from highest to lowest scaled distinction centrality. Node 8 ranks highest in both scaled distinction and status, but immediately afterward, Node 9 and Node 6 switch places in the ordering. Node 6 has the second-highest status, but the third-highest scaled distinction, while Node 9 goes from third in status to second in scaled distinction. The measure rewards the independent connections Node 9 has with Nodes 13 and 14 over the heavier constraints imposed on Node 6. [8] Node 10's relative position in the social hierarchy is consistent at fourth in both the status and scaled distinction measures.

The heterodoxy of Node 5 is elevated to the fifth position in the scaled distinction hierarchy, compared to the sixth position it holds in the status hierarchy. Node 7 correspondingly falls into the sixth position of the scaled distinction hierarchy from the fifth position in terms of status. However, scaled distinction centrality also elevates Nodes 1, 2, 3, and 4, which buttress 5's popularity. These nodes have the lowest status, yet leap over the hangers onto the core members of the main orthodox cluster. They are greatly penalized for their high level of constraint, whereas Node's 1, 2, 3, and 4 are empowered by 5's own lower status position. If the goal is to elevate uniqueness, this makes sense, as Nodes 11 through 16 are likely to conform to their higher-status, high-distinction connections. Nodes 7, 8, 9, and 10 already implicitly represent their contributions. Additionally, the application of the scalar is important in the final results. Without the scalar, Node 5 would actually be the third-ranked node in terms of distinction, leap-

[8] In the absence of Node 8's and its constraint effect on Node 6, Node 6 would become the most distinct and highest status.

Plots/search.png

(a) Reference Network Example

Figure 7: Example of a non-directed network of references, e.g. citation network, internet links, etc.

Table 7

Node	Distinction	SDistinction	Status	Constraint	Scalar
8	0.4939	1.0039	1.0000	0.5061	1.5099
9	0.3265	0.7680	0.8657	0.5392	1.5099
6	0.1530	0.6442	0.9633	0.8103	1.5099
10	0.1216	0.5367	0.8141	0.6925	1.5099
5	0.1837	0.3675	0.3603	0.1766	1.5099
7	-0.0898	0.1983	0.5649	0.6547	1.5099
1, 2, 3, 4	-0.2298	-0.1807	0.0963	0.3260	1.5099
16	-0.3764	-0.2994	0.1510	0.5273	1.5099
15	-0.5459	-0.4349	0.2175	0.7634	1.5099
13, 14	-0.5780	-0.4601	0.2313	0.8094	1.5099
11, 12	-0.707	-0.5707	0.2672	0.9742	1.5099

ing Node 6. Such a result may be desirable in technical applications where uniqueness should be privileged over social hierarchy. On the whole, the scaled distinction centrality measure helps highlight a node's uniqueness in search applications.

6 Implications for Theories of Distinction

6.1 Long Ties and Opportunities for Distinction

Given that networks are not static but evolve, the incentives to maximize distinction within a network can inform how culture evolves in response to network reorganization. As cultural capital is an outcome of both status and uniqueness, the same calculus of optimizing network position can be extended.

Hearkening to the stereotypical model of a village society, we can imagine a social world composed of dense local clusters interconnected only through elite networks. In such a network, most individuals would have dense ties within their local subgroup but few, if any, ties outside it. Such extra-local relationships would be managed by a handful of elites who would derive their status from the deference paid to them locally and through their brokerage with the wider world (see Tilly 1978 on the brokerage role of French priests and the nobility in France prior to the Revolution). The incentives for distinguishing oneself within a dense local subgroup through local stratification were likely to lead to a decline in the actor's status. In a tightly connected subgroup, stratification is low, and so the ability to differentiate is also low. Any initial actions are likely to lead to a loss of status greater than the gain in autonomy.

With these initial network conditions, the network pressures for cultural innovation are low. However, as economic and technological change brings actors into greater contact, the network of positions also undergoes transformation. The changes also begin to create greater local stratification. These changes create an opportunity structure for actors to create new connections in the network to increase their own distinction.

We would expect to see, in large datasets, a tendency for people who are locally constrained and of middling status to seek to reduce their constraint by forming long ties, because this can improve both their status and their autonomy. Those of high status might not jeopardize their distinction by associating with those of lower standing, while those of low status are constrained by others' reluctance to associate with them. The expected result would be a flourishing of long-tie formation in the middle levels of the distinction hierarchy. This prediction can be tested against large longitudinal, well-circumscribed networks, such as those in (**?**), in the future.

The insight that it is the middling levels that are most likely to form long ties to create their own distinction indicates that it is structures with a degree of stratification and the opportunity to form long ties that are most likely to create cultural heterogeneity, or, in other words, a flourishing of subcultures. Colleges are not merely opportunities for people from many backgrounds to recombine their cultural repertoires, but rather network structures that encourage the pursuit of cultural distinction, given the long ties that accompany students' new positions. The expansion of higher education beyond the old elites in the 1950s and 60s was a network transformation that gave rise to the major cultural shifts that began in these institutions. These cultural transformations were neither inherent in the students nor in the institutions, but arose from the network structure that enabled the pursuit of cultural distinction to match the newly acquired network position. Thus, if "liberals" are those who espouse diversity and innovation in cultural forms, to the extent that universities draw from a variety of social bases, liberal dispositions will result. However, if the university draws from an isolated basis, as the common educational destination of a delimited community, then the structural opportunities for cultural distinction are absent, and the culture is inherited from the isolated community.

The cultural practices of elites in a society are similarly dictated by the trade-offs between status and autonomy that arise from network structure. Bourdieu's French elite presents the ideal haute elite in terms of a stratified cultural disposition relative to lower-status members of society. However, the modern American elite has been characterized not by its sanctification of a limited set of cultural practices, but rather by its embrace of variety, or its cultural omnivorousness (**?**). The divergence between an elite interested in limited consecration and one interested in omnivorousness can be found in the network structure shaped by national geography and institutions. France, dominated by a single major capital as far back as the 1600s, had a singularly concentrated national elite that formed dense interconnections. The opportunity for enterprising distinction was foreclosed by interconnection. America's varied national elite geography, conversely, created locally cohesive elites that were connected less intensely at the national level. When brought into deeper integration after 1945, the elite themselves similarly formed long ties, creating the positional opportunity for the proliferation of elite cultural forms.

While the above exploration of how network shape changes diversity in cultural distinction practices is at least informative about how understanding the network of positions structures the cultural markers of distinction.

6.2 Capital Conversion as Network Integration of Fields

One of the problems with the relationship between multiple fields is how capital might be converted between fields. The "rate" between fields has always been a matter of confusion because it is difficult to quantify. Yet, it is an essential feature of an individual's navigation between the positional hierarchies of various fields. Seeking to convert a distinct position from one field to a better position in another is a critical strategy in the overall positional portfolio of agents.

However, when thought of as horizontally interlinked networks of vertical fields, the problem of capital conversion becomes tractable as a network problem. If someone from an economic field (broadly speaking) were to associate with someone high in some form of cultural capital, it refracts into the economic network as well because associations are at the level of the connections between individuals. Personal associations may be somewhat field-specific decisions, but unless field boundaries are rigorously enforced, associations between individuals are largely portfolio-level decisions.

Such a network position perspective aligns with the observation that those high in different types of capital are often willing to associate with one another, borrowing their mutual endorsements to advance their portfolios. Thus, those who are “low” in certain forms of capital can associate with those who are “high” in another type, insofar as it does not create undesirable network ties. The prototypical example is the artist and the benefactor. While the artist imparts cultural capital to the benefactor, the economic rewards, even if meager, are necessary for the artist. However, what is important for the benefactor is that the artist has the *right type* of low economic capital. If the economic capital of those at the height of the economy is built upon a workforce, associations with those who buttress their economic position would erode the overall portfolio. Such associations would close the distance that is necessary for having a distinct position.

Instead, one can associate with those who are low in economic capital *insofar* as they are not interconnected with the economic structures that buttress the position of the elites of the economic field. The artist connected to, or a part of, the proper working class undermines the structure of the economic field. The artist with a university education who may work a menial job is, through their social distance from the proper working class, does not erode the structure of the field. The conversion of cultural capital into economic capital is fundamentally about not having connections to the core structures of the economic field. Thus, those low in one type of capital and high in another will have the most success if the capital they lack is largely peripheral to the main structure of stratification in the field.

Note: ”the real logic of the functioning of capital [is the] conversion from one type to another” (1997,54) in Scott 238.

Similarly, those who are in distinct positions created by commonality in the individuals that structure the lower rungs of the network structure are able to easily transfer their position across fields. A sports star of low background being elevated in their social status by the working classes is no threat to the economic elites to affiliate freely with such an individual. The fields may differ, but they share a common scaffolding of working individuals.

Thus, the demarcation of the field becomes incredibly important in determining whether capital can be transferred. Within a field, practices become stratified, but when the field becomes defined as separate, capital among those most distinct in their fields can be exchanged if they share a common scaffolding. The key is the stratification rather than the cultural practice. Sports might be declasse from the perspective of cultural stratification, but the stratification is generally equivalent (general equivalence citation). Furthermore, stratifying the consumption experience is another way to make the “same” cultural object maintain the boundaries of distinction.

Floor or box seats, or season tickets, reinforce distinction while allowing for common consumption. Similarly, pop culture can be popular because of this aspect of stratification, the pedestal on which a producer of culture is placed. Common consumption

Plots/PortfolioIntegration.png

(a) Model of Viable Portfolio Integration

Figure 8

is not equivalent to common performance. Common consumption has the elevation of the producer, but common production lacks stratification, and thus must be stratified through avoidance by the distinct. Stratified common consumption is acceptable to elites, but un-stratified common consumption is to be avoided. However, those whose economic position may more closely match those who build up the scaffold must avoid common consumption because it would push their position into being coincident with those whom they are seeking to differentiate themselves from. This explains the avoidance of those who rely on cultural distinction for their position from popular cultural forms. At the same time, those who derive their capital from other fields are able to freely associate with popular cultural productions (which are stratified fields) while eschewing practices (which are un-stratified habitus within the cultural field). Capital conversion is not a mechanism of exchange conversion, but network integration. Understanding it as integration becomes theoretically generative.

6.3 Limitations on Positional Agency

One problem with the above measure is that is creates the spectre of limitless positional agency: if a connection is mutually distinction increasing, actors should pursue it. Given that distinction increasing positional changes are possible, but go un-enacted, there must be constraints to distinction maximizing action. Theoretical considerations must be made to account for costs that constrict connection formation for distinction optimization.

First, there is the matter of social foci (**?**), or the physical and social spaces that create co-location for the formation of connections as an *opportunity structure.* Brokers are particularly effective when they connect different clusters that would not otherwise be connected (Burt 1992), yet their advantaged position is otherwise threatened if the clusters begin to more directly connect. If the distance between them is maintained by the lack of overlap in their social foci, then re-positionings are foreclosed.

Second, there is the problem of *bargaining power.* The losers of positional change can retaliate by withdrawing their own connection to the changed node. This is particularly effective when multiple nodes withdraw their support in conjunction, with the retaliation leading the changed node to suffer a net decrease in network distinction. This runs counter to the predictions that would otherwise emerge from principles of triadic balance, as the maintenance of relational fragmentation is a legitimate strategy for producing high social distinction. Those who have experienced secondary education may be able to relate. Under conditions of fragmentation, where sides must be chosen, there is less of an issue of counter retaliation.

Third is the matter of *cognitive load.* Humans have a limited cognitive capacity and time to successfully maintain the commitments requisite of social relationships (**?**). While data may capture the structure of a specific field, people are part of many fields that remain unobserved in many empirical studies, but take away resources on their ability to act within the field of concern.

Fourth, is the *cross linkages of fields* in that re-positionings in one field can come at a cost in another field because while the fields are separated in their boundaries, they are interconnected at the level of the individual. The formation of a business connection with someone could create positional ramifications in the field of politics for example, and so positional change is otherwise unenacted. For example, in the case of the Medici, the economic ties that they formed with the new families had political implications that otherwise prevented the old families from forming similar connections because it would hurt their standing. It was only the Medici's earlier fall from grace that enabled such a toleration *padgett1993robust*.

Fifth is *positional uncertainty* about another's position in the field. While people are remarkably adept at inferring position in a field through the observation of habitus and capital resources, there remains a good deal of uncertainty about how new connections may upset existing competition structures, particularly in environments that are more circumscribed such as schools or organizations.

Sixth, differences in *habitus* mean that potential connections can fail to form because of the impacts it would have on their overall portfolio of positions. Just because of a potential gain in one area does not mean that it would not have costs in other areas that are unobserved. This is the problem above where capital conversion is not quite conversion, but incorporation.

While it is difficult to build such impediments to positional agency into the distinction measure, study designs can carefully control for the problems they may otherwise impose. Studies of societal conflict which aggregates across fields and within constrained social units can significantly alleviate limitations on positional agency that are external to the distinction measure.

7 Conclusion

Above we have introduced a way of constructing a general measure of distinction from network structure. The measure theorizes social distinction as a joint function of status and constraint derived from the pattern of network relations.

In considering how social actors make decisions to maximize their social distinction within a network, scholars must consider the network counterfactuals that actors themselves weigh. In the case of the Medici network, actors considered their position within a network both with the Medici and without the Medici, and those who would increase their social distinction in the Medici's absence undertook action to remove them. In the case of the karate network, the decision of the network to split was not made by Node 9, but 9 was forced to decide which pattern of relations to maintain. Imagining Node 9 in each of the two networks, 9's social distinction was higher in Mr. Hi's group than in the officer's group, and so 9 chose this more desirable place in the social world.

None of this is to suggest that actors are able to do the macro-level network calculation of distinction for the entire network when making decisions about their pattern of relations. However, humans are remarkable good at inputting in the network picture despite lacking the particulars and intuiting their position and optimal moves. Exploring the cognitive shortcuts that people use to map the complex details of their surrounding networks is an area ripe for academic inquiry. In addition, there is a future where computer modeling can estimate network evolution through individual maximization.

This paper simply seeks to establish a measure of social distinction within a network structure. While the examples have focused on social distinction in shaping group politics, it is hoped that the measure proposed can be applied to and expand understandings of all forms of social distinction. Within sociology, the measure may prove particularly useful for developing new theory and explanations for shifts in competition over cultural distinction. However, use cases are anticipated to expand beyond the social world. The benefit of the measure maintains the rank of the highest status nodes within a system, but elevates those nodes that derive their position from non-redundant nomination sources. In the social world of power, politics, and alliances, it rewards the independence of one's support. In a task such as ordering search results, it increases the diversity of information that would be selected in the generation of a ranked list, eliminating redundant information from central cliques of high-status actors. The measure may hopefully find a wide variety of future applications.

Appendix: Distinction Centrality Function

```
distinction <- function(x, norm = "abm", digits = 4) {
  n <- vcount(x)

  # Ensure vertices have sequential numeric names for indexing, preserving
      original labels
  if (is.null(V(x)$name)) {
    V(x)$name <- as.character(1:n)
    names <- V(x)$name
  } else {
    names <- V(x)$name
    V(x)$name <- as.character(1:n)
  }

  # Normalization functions for status (eigenvector centrality) scores
  # Handled division by zero for sparse/empty subgraphs
  apply_norm <- function(v, norm_type) {
    if (norm_type == "max") {
      if (max(v^2) == 0) v else v^2 / max(v^2)
    } else if (norm_type == "one") {
      v^2
    } else if (norm_type == "abm") {
      if (max(abs(v)) == 0) v else abs(v) / max(abs(v))
    } else if (norm_type == "abs") {
      abs(v)
    } else {
      v
    }
  }

  # Helper to compute status for any graph (handles components and
     isolates robustly)
  get_status <- function(g) {
    nv <- vcount(g)
    s_vec <- numeric(nv)
    xc <- components(g)
    # Process each component independently to avoid global eigen problems
    for (k in 1:xc$no) {
      nodes_in_comp <- which(xc$membership == k)
      if (length(nodes_in_comp) > 1) {
        subg <- induced_subgraph(g, nodes_in_comp)
        # Replaced base::eigen with igraph::eigen_centrality (ARPACK) for
            O(E) efficiency instead of O(V^3)
        eig <- suppressWarnings(eigen_centrality(subg, scale = FALSE)$
            vector)
        s_vec[nodes_in_comp] <- abs(eig)
      } else {
        s_vec[nodes_in_comp] <- 0
      }
```

```
  }
  return(s_vec)
}

# 1. Compute main graph status (s)
s <- get_status(x)

# 2. Normalize main status
s <- apply_norm(s, norm)

d <- numeric(n) # Distinction vector
u <- numeric(n) # Constraint vector (average neighbor status in deleted
   subgraph)

# 3. Iterate through each node to calculate induced constraint (
   jackknife-like approach)
for (i in 1:n) {
  j <- neighbors(x, i)

  # Optimization: Skip subgraph eigen decomposition if the focal node
     has no neighbors
  if (length(j) == 0) {
    u[i] <- 0
    d[i] <- s[i]
    next
  }

  xd <- delete_vertices(x, i)

  # Compute status of neighbors in the node-deleted subgraph (xd)
  # Fixed boolean coercion bug (was checking if FALSE == 0)
  if (ecount(xd) > 0 && vcount(xd) > 1) {
    sd <- get_status(xd)
    # Re-apply same normalization to deleted subgraph status
    sd <- apply_norm(sd, norm)
  } else {
    sd <- rep(0, vcount(xd))
  }

  names(sd) <- V(xd)$name
  # Constraint (u) = mean status of focal node's neighbors in deleted
     graph
  # Robust indexing using as_ids() to fetch original names
  u[i] <- mean(sd[as.character(as_ids(j))])
  d[i] <- s[i] - u[i] # Distinction (d) = Status - Constraint
}

# 4. Final aggregation and scalar-adjusted distinction (scd) calculation
d[is.na(d)] <- 0
u[is.na(u)] <- 0
```

```
  # Handled division by zero if entire network has 0 status
  mean_s <- mean(s)
  scalar <- if (mean_s == 0) 0 else mean(u) / mean_s
  scd <- (s * scalar) - u

  # Assemble output dataframe
  dat <- data.frame(d = d, scd = scd, s = s, u = u, scalar = scalar)
  dat <- round(dat, digits)
  dat <- data.frame(n = names, dat)
  rownames(dat) <- 1:nrow(dat)
  return(dat)
}
```